

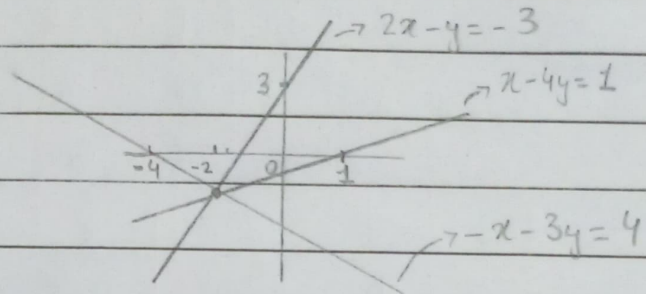
LA-Assignment 1

Q1)

$$x - 4y = 1$$

$$2x - y = -3$$

$$-x - 3y = 4$$



$$\begin{bmatrix} 1 & -4 & | & 1 \\ 2 & -1 & | & -3 \\ -1 & -3 & | & 4 \end{bmatrix}$$

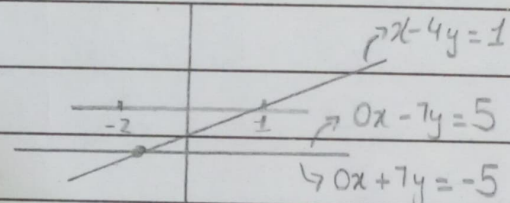
pivot $\leftarrow$

$$\begin{bmatrix} 1 & -4 & | & 1 \\ 0 & 7 & | & -5 \\ 0 & -7 & | & 5 \end{bmatrix}$$

replacement

$$R_2 - 2R_1$$

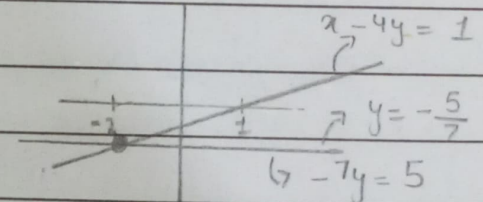
$$R_3 + R_1$$



$$\begin{bmatrix} 1 & -4 & | & 1 \\ 0 & 1 & | & -5/7 \\ 0 & -7 & | & 5 \end{bmatrix}$$

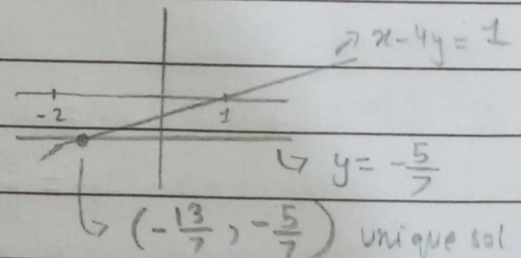
scaling

$$\frac{1}{7} R_2$$



$$\begin{bmatrix} 1 & -4 & | & 1 \\ 0 & 1 & | & -5/7 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$R_3 + 7R_2$$



so

$$x - 4y = 1$$

$$y = -5/7$$

substitute y value

$$x - 4(-5/7) = 1$$

$$x = 1 - 20/7$$

$$x = -13/7$$

$$\therefore \left(-\frac{13}{7}, -\frac{5}{7}\right)$$

All three lines meet at exactly one point $\left(-\frac{13}{7}, -\frac{5}{7}\right)$. This shows the system is consistent and has a unique solution.

Q2)

$$\text{pivot} \rightarrow \begin{bmatrix} 1 & -4 & 7 & | & g \\ 0 & 3 & -5 & | & h \\ -2 & 5 & -9 & | & k \end{bmatrix}$$

$$\text{pivot} \rightarrow \begin{bmatrix} 1 & -4 & 7 & | & g \\ 0 & 1 & -5/3 & | & h/3 \\ 0 & -3 & 5 & | & k+2g \end{bmatrix} \quad \begin{array}{l} 1/3 R_2 \\ R_3 + 2R_1 \end{array} \quad \begin{array}{l} \text{scaling} \\ \text{replacement} \end{array}$$

$$\begin{bmatrix} 1 & -4 & 7 & | & g \\ 0 & 1 & -5/3 & | & h/3 \\ 0 & 0 & 0 & | & k+2g+h \end{bmatrix} \quad R_3 + 3R_2$$

$$0 = k + 2g + h \rightarrow h + k + 2g = 0$$

For the system to be consistent, any all zero ^{left} ~~right~~ hand row must have a zero on the right as well

$$\cancel{h = -k - 2g}; \cancel{k = -h - 2g} \quad g = \frac{1}{2}(-h - k)$$

$\therefore$ if $h + k + 2g \neq 0$ the system is inconsistent (no solution)
but when $h + k + 2g = 0$, the system is consistent and has infinitely many solutions. (or when $h = -k - 2g$ it is $0 = 0$ so consistent)

Q3)

$$w + x + y + z = 6$$

$$w + 0 + y + z = 4$$

$$w + 0 + y + 0 = 2$$

$$a) \begin{array}{c} w \quad x \quad y \quad z \quad b \\ \begin{bmatrix} 1 & 1 & 1 & 1 & | & 6 \\ 1 & 0 & 1 & 1 & | & 4 \\ 1 & 0 & 1 & 0 & | & 2 \end{bmatrix} \end{array}$$

apply row operations

$$\text{pivot} \leftarrow \begin{bmatrix} \underline{1} & 1 & 1 & 1 & | & 6 \\ 0 & -1 & 0 & 0 & | & -2 \\ 0 & -1 & 0 & -1 & | & -4 \end{bmatrix}$$

replacement
 $R_2 - R_1$
 $R_3 - R_1$

$$\text{pivot} \leftarrow \begin{bmatrix} \underline{1} & 1 & 1 & 1 & | & 6 \\ 0 & \underline{1} & 0 & 0 & | & 2 \\ 0 & -1 & 0 & -1 & | & -4 \end{bmatrix}$$

Scaling
 $-R_2$

$$\begin{bmatrix} \underline{1} & 1 & 1 & 1 & | & 6 \\ 0 & 1 & 0 & 0 & | & 2 \\ 0 & 0 & 0 & -1 & | & -2 \end{bmatrix}$$

replacement
 $R_2 + R_3$

$$\begin{bmatrix} \underline{1} & 1 & 1 & 1 & | & 6 \\ 0 & \underline{1} & 0 & 0 & | & 2 \\ 0 & 0 & 0 & \underline{1} & | & 2 \end{bmatrix}$$

Scaling
 $-R_3$

$$\begin{bmatrix} 1 & 1 & 1 & 0 & | & 4 \\ 0 & 1 & 0 & 0 & | & 2 \\ 0 & 0 & 0 & \underline{1} & | & 2 \end{bmatrix}$$

replacement
 $R_1 - R_3$

$$\begin{bmatrix} \underline{1} & 0 & 1 & 0 & | & 2 \\ 0 & \underline{1} & 0 & 0 & | & 2 \\ 0 & 0 & 0 & \underline{1} & | & 2 \end{bmatrix}$$

replacement
 $R_1 - R_2$

$$\begin{bmatrix} 1 & 0 & 1 & 0 & | & 2 \\ 0 & 1 & 0 & 0 & | & 2 \\ 0 & 0 & 0 & \underline{1} & | & 2 \end{bmatrix}$$

pivots

RREF being done

a) Leading Variable (pivot) = w, x, z b) Free Variable = y

c) General solution:

let $y = t$ where $t \in \mathbb{R}$

$$\therefore w + y = 2 \quad ; \quad x = 2 \quad ; \quad z = 2$$

$$w + t = 2$$

$$w = 2 - t$$

$$d) \left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 2 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 2 \\ -2 & 0 & -2 & 0 & -3 \end{array} \right]$$

$$\text{added } -2w - 2y = -3$$

doing row operation

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 2 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 \end{array} \right]$$

replacement

$$R_4 + 2R_1$$

as

Row 4 is $0 = 1$ which is not valid for a system to be consistent. $0 \neq 1$.

Hence, solution with the fourth equation is Inconsistent
(No solution)

Q4a) Solving System 1 using Gauss Jordan Method

$$x + y + z = 6$$

$$x + 2y + 2z = 11$$

$$2x + 3y - 4z = 3$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 2 & 11 \\ 2 & 3 & -4 & 3 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 5 \\ 0 & 1 & -6 & -9 \end{array} \right]$$

Replacement

$$R_2 - R_1$$

$$R_3 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & -7 & -14 \end{array} \right] \rightarrow \text{replacement}$$

$$R_3 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 1 & 2 \end{array} \right] \rightarrow \text{scaling}$$

$$-\frac{1}{7} R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 1 & 2 \end{array} \right] \rightarrow R_1 - R_2 \quad \text{replacement}$$

$$\cancel{R_2} \quad \cancel{R_3}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{array} \right] \rightarrow \text{replacement}$$

$$R_2 - R_3$$

$$\therefore \text{The RREF of System is } \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

Now solving system 2 using Gauss Jordan Method

$$x + y + z = 7$$

$$x + 2y + 2z = 10$$

$$2x + 3y - 4z = 3$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 7 \\ 1 & 2 & 2 & 10 \\ 2 & 3 & -4 & 3 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 7 \\ 0 & 1 & 1 & 3 \\ 0 & 1 & -6 & -11 \end{array} \right] \rightarrow \text{replacement}$$

$$R_2 - R_1$$

$$R_3 - 2R_1$$

$$\left[\begin{array}{ccc|c} \underline{1} & 1 & 1 & 7 \\ 0 & \underline{1} & 1 & 3 \\ 0 & 0 & -7 & -14 \end{array} \right] \quad \text{replacement}$$

$$R_3 - R_2$$

$$\left[\begin{array}{ccc|c} \underline{1} & 1 & 1 & 7 \\ 0 & \underline{1} & 1 & 3 \\ 0 & 0 & \underline{1} & 2 \end{array} \right] \quad \text{scaling}$$

$$-\frac{1}{7} R_3$$

$$\left[\begin{array}{ccc|c} \underline{1} & 0 & 0 & 4 \\ 0 & \underline{1} & 1 & 3 \\ 0 & 0 & \underline{1} & 2 \end{array} \right] \quad \begin{array}{l} R_1 - R_2 \\ \text{replacement} \end{array}$$

$$\left[\begin{array}{ccc|c} \underline{1} & 0 & 0 & 4 \\ 0 & \underline{1} & 0 & 1 \\ 0 & 0 & \underline{1} & 2 \end{array} \right] \quad \begin{array}{l} \text{replacement} \\ R_2 - R_3 \end{array}$$

∴ The RREF of System 2 is

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

Q4b) Solving System 1 using inversion Algorithm

$$x + y + z = 6$$

$$x + 2y + 2z = 11$$

$$2x + 3y - 4z = 3$$

$$[A | I] \xrightarrow{\text{row operation}} [I | A^{-1}]$$

$$\left[\begin{array}{ccc|ccc} \underline{1} & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 2 & 0 & 1 & 0 \\ 2 & 3 & -4 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} \underline{1} & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & -6 & -2 & 0 & 1 \end{array} \right] \quad \begin{array}{l} \text{replacement} \\ R_2 - R_1 \\ R_3 - 2R_1 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & -7 & -1 & -1 & 1 \end{array} \right] \quad \text{replacement}$$

 $R_3 - R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & \frac{1}{7} & \frac{1}{7} & -\frac{1}{7} \end{array} \right] \quad \text{scaling}$$

 $-\frac{1}{7} R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -1 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & \frac{1}{7} & \frac{1}{7} & -\frac{1}{7} \end{array} \right] \quad \begin{array}{l} R_1 - R_2 \\ \text{replacement} \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -1 & 0 \\ 0 & 1 & 0 & -8/7 & 6/7 & 1/7 \\ 0 & 0 & 1 & 1/7 & 1/7 & -1/7 \end{array} \right] \quad R_2 - R_3$$

$$\therefore \text{Thus } A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ -8/7 & 6/7 & 1/7 \\ 1/7 & 1/7 & -1/7 \end{bmatrix}$$

now solving the system

$$x = A^{-1} \{b\}$$

$$x = \begin{bmatrix} 2 & -1 & 0 \\ -8/7 & 6/7 & 1/7 \\ 1/7 & 1/7 & -1/7 \end{bmatrix} \cdot \begin{bmatrix} 6 \\ 11 \\ 3 \end{bmatrix}$$

$$\rightarrow \downarrow$$

$$12 - 11 + 0$$

$$x = \begin{bmatrix} 2(6) + (11)(-1) + 0(3) \\ -8/7(6) + 6/7(11) + 1/7(3) \\ 1/7(6) + 1/7(11) - 1/7(3) \end{bmatrix}$$

$$x = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$$

 $\rightarrow$ solution of system 1

Solving System 2 using inversion algorithm

$$x + y + z = 7$$

$$x + 2y + 2z = 10$$

$$2x + 3y - 4z = 3$$

$$[A | I] \xrightarrow{\text{row operations}} [I | A^{-1}]$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 2 & 0 & 1 & 0 \\ 2 & 3 & -4 & 0 & 0 & 1 \end{array} \right]$$

Note the $[A | I]$ expression matches system 1 form so I am writing the result directly as the steps are same

$$\therefore [I | A^{-1}] = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -1 & 0 \\ 0 & 1 & 0 & -8/7 & 6/7 & 1/7 \\ 0 & 0 & 1 & 1/7 & 1/7 & -1/7 \end{array} \right]$$

$$\therefore A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ -8/7 & 6/7 & 1/7 \\ 1/7 & 1/7 & -1/7 \end{bmatrix}$$

now solving system

$$x = A^{-1}b$$

$$x = \begin{bmatrix} 2 & -1 & 0 \\ -8/7 & 6/7 & 1/7 \\ 1/7 & 1/7 & -1/7 \end{bmatrix} \begin{bmatrix} 7 \\ 10 \\ 3 \end{bmatrix}$$

$$x = \begin{bmatrix} 2(7) + (-1)(10) + 0(3) \\ -8/7(7) + 6/7(10) + 1/7(3) \\ 1/7(7) + 1/7(10) - 1/7(3) \end{bmatrix}$$

$$x = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

→ solution of system 2

Q5)

$$C = \begin{bmatrix} a & 1 & 0 \\ 0 & a & 1 \\ 1 & 0 & a \end{bmatrix}$$

$$5a) |C| = a \begin{vmatrix} a & 1 \\ 0 & a \end{vmatrix} - 1 \begin{vmatrix} 0 & 1 \\ 1 & a \end{vmatrix} + 0 \begin{vmatrix} 0 & a \\ 1 & 0 \end{vmatrix}$$

$$|C| = a(a^2 - 0) - 1(0 - 1) + 0$$

$$|C| = a^3 + 1$$

5b) A square matrix (C) is invertible if and only if its determinant is not zero

therefore C is invertible exactly when

$$a^3 + 1 \neq 0$$

$$\sqrt[3]{a^3} \neq \sqrt[3]{-1}$$

$$a \neq -1$$

Hence C is invertible for all real $a \neq -1$

c) Consider the linear system $Cx = b$, where C is coefficient matrix

→ If $a \neq -1$ then $\det(C) \neq 0$ so C is invertible. Therefore the system has unique solution for any value of b.

→ if $a = -1$ then $\det(C) = 0$ so C is singular. Therefore the system may have infinitely many solution, or no solutions depending on b.

Let's solve when $a = -1$ $\therefore C = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{bmatrix}$

Augmented Matrix for $Cx = b$ when $b = [b_1, b_2, b_3]$

$$\left[\begin{array}{ccc|c} -1 & 1 & 0 & b_1 \\ 0 & -1 & 1 & b_2 \\ 1 & 0 & -1 & b_3 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & -b_1 \\ 0 & -1 & 1 & b_2 \\ 1 & 0 & -1 & b_3 \end{array} \right] \quad \text{scaling} = R_1$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & -b_1 \\ 0 & 1 & -1 & -b_2 \\ 0 & -1 & -1 & b_3 + b_1 \end{array} \right] \quad \begin{array}{l} -R_2 \\ R_3 - R_1 \end{array} \quad \begin{array}{l} \text{scaling} \\ \text{replacement} \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & -b_1 \\ 0 & 1 & -1 & -b_2 \\ 0 & 0 & 0 & b_3 + b_1 + b_2 \end{array} \right] \quad R_3 - R_2$$

so for consistency we require
 $b_1 + b_2 + b_3 = 0$

$\therefore$ For $a = -1$

if $b_1 + b_2 + b_3 = 0$, we have infinitely many solution
 if $b_1 + b_2 + b_3 \neq 0$, we have no solution

Q6)

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x, y) = (x + 2y, y)$$

a) Matrix Representation $\Rightarrow A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

b) Determinant $(A) = ad - bc = 1 - 0 = 1$ so it means transformation area preserves and orientation preserving (this rules out reflection)
 The eigenvalues of A are both 1 (so this rules out rotation)

It is not scaling either as A is not diagonal, it has a non zero off diagonal element.

Now a sheer matrix is in form $\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$ which matches

matrix A with $k=2$.

Geometric behavior: for any point (x, y) , $T(x, y) = (x + 2y, y)$
the y -coordinate remains unchanged and x -coordinate is shifted by $2y$.

if $y > 0$, shift right

if $y < 0$, shift left

if $y = 0$, no shift

$\therefore$ Matrix A is a sheer.

Q6c) Let square have corners

$P(0,0)$ $Q(1,0)$ $R(1,1)$ $S(0,1)$

$$T(P) = (0,0)$$

$$T(x) = (x + 2y, y)$$

$$0 + 0, 0 = 0, 0$$

$$1 + 0, 0 = 1, 0$$

$$1 + 2, 1 = 3, 1$$

$$0 + 2, 1 = 2, 1$$

$$T(Q) = (1,0)$$

$$T(R) = (3,1)$$

$$T(S) = (2,1)$$

This applying once sheared a square into a parallelogram.
Applying repeatedly would cause

$$T^n(x, y) = (x + 2ny, y)$$

$$T^n(P) = (0,0)$$

$$T^n(Q) = (1,0)$$

$$T^n(R) = (1 + 2n, 1)$$

$$T^n(S) = (2n, 1)$$

Q7)

$$\begin{aligned}ax + by - 3z &= -3 \\ -2x - by + cz &= -1 \\ ax + 3y - cz &= -3\end{aligned}$$

$$x=1, y=-1 \text{ and } z=2$$

Substitute in eq

$$~~a~~ a - b - 6 = -3$$

$$a - b = 3 \quad \rightarrow \text{eq 1}$$

$$-2 + b + 2c = -1$$

$$b + 2c = 1 \quad \rightarrow \text{eq 2}$$

$$a - 3 - 2c = -3$$

$$a - 2c = 0 \quad \rightarrow \text{eq 3}$$

$$a - b = 3 \quad \text{eq 1}$$

$$b + 2c = 1 \quad \text{eq 2}$$

$$a - 2c = 0 \quad \text{eq 3}$$

solve

$$\begin{array}{l} a - 2c = 0 \\ a = 2c \end{array} \rightarrow \begin{array}{l} a - b = 3 \\ 2c - b = 3 \\ 2c = 3 + b \end{array} \rightarrow \begin{array}{l} b + 2c = 1 \\ b + (3 + b) = 1 \\ 2b = -2 \\ b = -1 \end{array}$$

for c	for a
$2c = 3 + b$	$a = 2c$
$2c = 3 - 1$	$a = 2(1)$
$c = 1$	$a = 1$

$$\therefore a = 1, b = -1, c = 1 \quad \text{Ans}$$

$$\begin{aligned} \text{Verify} \rightarrow 2(1) + (-1)(-1) - 3(2) &\equiv -3 \\ -3 &\equiv -3 \quad \text{proof} \end{aligned}$$

$$\begin{aligned} -2(1) - (-1)(-1) + 1(2) &\equiv -1 \\ -1 &\equiv -1 \quad \text{proof} \end{aligned}$$

$$\begin{aligned} 2(1) + 3(-1) - 1(2) &\equiv -3 \\ -3 &\equiv -3 \quad \text{proof} \end{aligned}$$

Q8) If the coefficient matrix has a pivot in every row, then the REF form of the augmented matrix has no inconsistent equation (no row in the form $[0 \ 0 \ \dots \ 0 \ | \ d]$ with $d \neq 0$). Therefore, having pivot in each row prevents this and keeps the system consistent.

Q9a)

Matrix

Date: _____

~~Q9a)~~ ~~Matrix~~ $A = \begin{bmatrix} 5 & 3 & 2 \\ 3 & 3 & 2 \\ 0 & 1 & 1 \end{bmatrix}$

 $[A | I] \xrightarrow{\text{row operation}} [I | A^{-1}]$

$$\left[\begin{array}{ccc|ccc} 5 & 3 & 2 & 1 & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 1 & \frac{3}{5} & \frac{2}{5} & \frac{1}{5} & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \quad \frac{1}{5} R_1 \text{ scaling}$$

$$\left[\begin{array}{ccc|ccc} 1 & \frac{3}{5} & \frac{2}{5} & \frac{1}{5} & 0 & 0 \\ 0 & \frac{6}{5} & \frac{4}{5} & -\frac{3}{5} & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \quad \begin{array}{l} \text{replacement} \\ R_2 - 3R_1 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & \frac{3}{5} & \frac{2}{5} & \frac{1}{5} & 0 & 0 \\ 0 & 1 & \frac{2}{3} & -\frac{1}{2} & \frac{5}{6} & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \quad \begin{array}{l} \text{scaling} \\ \frac{5}{6} R_2 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & \frac{3}{5} & \frac{2}{5} & \frac{1}{5} & 0 & 0 \\ 0 & 1 & \frac{2}{3} & -\frac{1}{2} & \frac{5}{6} & 0 \\ 0 & 0 & \frac{1}{3} & \frac{1}{2} & -\frac{5}{6} & 1 \end{array} \right] \quad \begin{array}{l} \text{replacement} \\ R_3 - R_2 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & \frac{3}{5} & \frac{2}{5} & \frac{1}{5} & 0 & 0 \\ 0 & 1 & \frac{2}{3} & -\frac{1}{2} & \frac{5}{6} & 0 \\ 0 & 0 & 1 & \frac{3}{2} & -\frac{5}{2} & 3 \end{array} \right] \quad \begin{array}{l} \text{scaling} \\ 3R_3 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & \frac{2}{3} & -\frac{1}{2} & \frac{5}{6} & 0 \\ 0 & 0 & 1 & \frac{3}{2} & -\frac{5}{2} & 3 \end{array} \right] \quad \begin{array}{l} R_1 - \frac{3}{5} R_2 \\ R_1 - \frac{3}{5} R_2 \text{ replacement} \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & 0 & -\frac{3}{2} & \frac{5}{2} & -2 \\ 0 & 0 & 1 & \frac{3}{2} & -\frac{5}{2} & 3 \end{array} \right] \quad R_2 - \frac{2}{3} R_3$$

$$\text{Thus } A^{-1} = \begin{bmatrix} 1/2 & -1/2 & 0 \\ -3/2 & 5/2 & -2 \\ 3/2 & -5/2 & 3 \end{bmatrix}$$

Verifying AA^{-1}

$$AA^{-1} = I$$

$$\begin{bmatrix} 5 & 3 & 2 \\ 3 & 3 & 2 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1/2 & -1/2 & 0 \\ -3/2 & 5/2 & -2 \\ 3/2 & -5/2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 5(0.5) + 3(-1.5) + 2(1.5) & 5(-0.5) + 3(2.5) + 2(-2.5) & 5(0) + 3(-2) + 2(3) \\ 3(0.5) + 3(-1.5) + 2(1.5) & 3(-0.5) + 3(2.5) + 2(-2.5) & 3(0) + 3(-2) + 2(3) \\ 0(0.5) + 1(-1.5) + 1(1.5) & 0(-0.5) + 1(2.5) + 1(-2.5) & 0(3) + 1(-2) + 1(3) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

hence verified that for system 1 we correctly calculated the inverse

Now Solving Matrix 2

$$B = \begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix}$$

$$[A | I] \xrightarrow{\text{row operation}} [I | A^{-1}]$$

$$\left[\begin{array}{ccc|ccc} -1 & 3 & -4 & 1 & 0 & 0 \\ 2 & 4 & 1 & 0 & 1 & 0 \\ -4 & 2 & -9 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 1 & -3 & 4 & -1 & 0 & 0 \\ 2 & 4 & 1 & 0 & 1 & 0 \\ -4 & 2 & -9 & 0 & 0 & 1 \end{array} \right] \begin{array}{l} -R_1 \\ \text{scaling} \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & -3 & 4 & -1 & 0 & 0 \\ 0 & 10 & -7 & 2 & 1 & 0 \\ 0 & -10 & 7 & -4 & 0 & 1 \end{array} \right] \begin{array}{l} \text{replacement} \\ R_2 - 2R_1 \\ R_3 + 4R_1 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & -3 & 4 & -1 & 0 & 0 \\ 0 & 10 & -7 & 2 & 1 & 0 \\ 0 & 0 & 0 & -2 & 1 & 1 \end{array} \right] \begin{array}{l} \text{scaling and replacement} \\ \frac{1}{10} R_2 \\ R_3 + 10R_2 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & -3 & 4 & -1 & 0 & 0 \\ 0 & 1 & -7/10 & 1/5 & 1/10 & 0 \\ 0 & 0 & 0 & -2 & 1 & 1 \end{array} \right]$$

Since row 3 consists solely of zeroes the determinant of the matrix equals 0. Thus the matrix is not invertible.

Q9b) For ~~sys~~ Matrix A

$$A = \begin{bmatrix} 5 & 3 & 2 \\ 3 & 3 & 2 \\ 0 & 1 & 1 \end{bmatrix}$$

$$Ax = 0$$

Since A is invertible, the only solution is the trivial solution.

$$x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Now solving matrix B

$$B = \begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix}$$

$$Bx = 0$$

$$\begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

as per part a the Row operation we applied and reduced the matrix I am picking from where I left out

row operation as per part a of second matrix gives this result.

$$\begin{bmatrix} 1 & -3 & 4 & | & 0 \\ 0 & 1 & -7/10 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

x y z b

~~$x - 3y + 4z = 0$; $y - 7/10 z = 0$; let $z = t$~~

Apply RREF

where $t \in \mathbb{R}$

substitute

$$\begin{bmatrix} 1 & -3 & 4 & | & 0 \\ 0 & 1 & -7/10 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 19/10 & | & 0 \\ 0 & 1 & -7/10 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$R_1 + 3R_2$ replacement

$$x + 19/10 z = 0 \quad ; \quad y - 7/10 z = 0 \quad ; \quad \text{let } z = t$$

where $t \in \mathbb{R}$

substitute

$$x + \left(\frac{19}{10}\right)t = 0 \quad ; \quad y - \left(\frac{7}{10}\right)t = 0 \quad ; \quad z = t$$

$$x = \frac{-19t}{10} \quad ; \quad y = \frac{7t}{10} \quad ; \quad z = t$$

Q9c) Solving Matrix A

$$A = \begin{bmatrix} 5 & 3 & 2 \\ 3 & 3 & 2 \\ 0 & 1 & 1 \end{bmatrix} \quad \cdot b = \begin{bmatrix} 4 \\ 2 \\ 5 \end{bmatrix}$$

$$x = A^{-1}b$$

$$x = \begin{bmatrix} 1/2 & -1/2 & 0 \\ -3/2 & 5/2 & -2 \\ 3/2 & -5/2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \\ 5 \end{bmatrix}$$

$$x = \begin{bmatrix} 4(0.5) + 2(-0.5) + 5(0) \\ 4(-1.5) + 2(2.5) + 5(-2) \\ 4(1.5) + 2(-2.5) + 5(3) \end{bmatrix}$$

$$x = \begin{bmatrix} 1 \\ -11 \\ 16 \end{bmatrix}$$

Answer

Solving matrix B

$$B = \begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 2 \\ 5 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} -1 & 3 & -4 & 4 \\ 2 & 4 & 1 & 2 \\ -4 & 2 & -9 & 5 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} \boxed{1} & -3 & 4 & -4 \\ 0 & 10 & -7 & +10 \\ 0 & -10 & 7 & -11 \end{array} \right] \quad \begin{array}{l} -R_1 \\ R_2 - 2R_1 \\ R_3 + 4R_1 \end{array} \quad \left. \begin{array}{l} \text{scaling and} \\ \text{replacement} \end{array} \right\}$$

$$\left[\begin{array}{ccc|c} 1 & -3 & 4 & -4 \\ 0 & 1 & -7_{10} & 1 \\ 0 & 0 & 0 & -1 \end{array} \right] \quad \left. \begin{array}{l} \frac{1}{10} R_2 \\ R_3 + 10 R_2 \end{array} \right\} \begin{array}{l} \text{scaling and} \\ \text{replacement} \end{array}$$

since the last row is $[0 \ 0 \ 0 \ | \ -1]$ which means $0 = -1$
therefore it is an inconsistency.

Hence there is no solution to $Bx = b$

$$\textcircled{211}) \quad u = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \quad b = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix} \quad c = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$$

$$T(x) = (x_1 - 3x_2, 3x_1 + 5x_2, -x_1 + 7x_2)$$

a) $\mathbb{R}^2 \rightarrow \mathbb{R}^3$ so $m=3$ $n=2$ it's a 3×2 matrix

$$A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$$

→ standard matrix

b) $T(u) = ?$ $u = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

$$T(u) = Au$$

so $T(u) = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

$$T(u) = \begin{bmatrix} 1(2) + (-3)(-1) \\ 3(2) + 5(-1) \\ -1(2) + 7(-1) \end{bmatrix}$$

$$T(u) = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$$

c) $T(x) = b$ where $b = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$

$$x \in \mathbb{R}^2$$

so ~~matrix~~ $A = \begin{bmatrix} \cancel{x_1} \\ \cancel{x_2} \end{bmatrix}$

$$Ax = b$$

$$A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = b$$

so linear system

$$x_1 - 3x_2 = 3$$

$$3x_1 + 5x_2 = 2$$

$$-x_1 + 7x_2 = -5$$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & -5 \end{array} \right]$$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & -2 \end{array} \right]$$

replacement

$$R_2 - 3R_1$$

$$R_3 + R_2$$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 4 & -2 \end{array} \right]$$

scaling

$$1/4 R_2$$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{array} \right]$$

replacement

$$R_3 - 4R_2$$

$$\left[\begin{array}{cc|c} 1 & 0 & 3/2 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{array} \right]$$

$$R_1 + 3R_2$$

replacement

so

$$x_1 = 3/2$$

$$x_2 = -1/2$$

Answer

$$\therefore x = (3/2, -1/2)$$

✱

d) No, there is exactly one x such that $T(x) = b$ as the above system is consistent and has unique solution

e) $T(x) = c$

so $x_1 - 3x_2 = 3$

$3x_1 + 5x_2 = 2$

$-x_1 + 7x_2 = 5$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{array} \right]$$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & 8 \end{array} \right] \begin{array}{l} \text{replacement} \\ R_2 - 3R_1 \\ R_3 + R_1 \end{array}$$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 4 & 8 \end{array} \right] \begin{array}{l} \text{scaling} \\ 1/4 R_2 \end{array}$$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & -7 \\ 0 & 0 & 10 \end{array} \right] \begin{array}{l} \text{replacement} \\ R_3 - 4R_2 \end{array}$$

as last row is $0=10$, which is inconsistent. Therefore no solution exists.

Hence c is not in range of T .

Q12) $T\left(\underset{\downarrow v_1}{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}\right) = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$, $T\left(\underset{\downarrow v_2}{\begin{bmatrix} 2 \\ 3 \end{bmatrix}}\right) = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$

Standard matrix $A = [T(e_1), T(e_2)]$ $e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + k_2 \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad , \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Solve e_1

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + k_2 \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} k_1 + 2k_2 \\ k_1 + 3k_2 \end{bmatrix}$$

So,

$$\begin{cases} k_1 + 2k_2 = 1 & \rightarrow \text{eq 1} \\ k_1 + 3k_2 = 0 & \rightarrow \text{eq 2} \end{cases}$$

Solve using substitution

$$k_1 + 2k_2 = 1$$

$$k_1 = -3k_2 \rightarrow \text{eq 3}$$

Substitute eq 3 in eq 1

$$-3k_2 + 2k_2 = 1$$

$$-1 = k_2$$

for k

$$k_1 = -3k_2$$

$$k_1 = -3(-1)$$

$$k_1 = 3$$

Solve e_2

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} c_1 + 2c_2 \\ c_1 + 3c_2 \end{bmatrix}$$

So,

$$c_1 + 2c_2 = 0 \rightarrow \text{eq 1}$$

$$c_1 + 3c_2 = 1 \rightarrow \text{eq 2}$$

~~eq 1~~ Solving using substitution

$$c_1 = -2c_2 \rightarrow \text{eq 3}$$

Substitute eq 3 in eq 2

$$\begin{aligned} c_1 + 3c_2 &= 1 \\ -2c_2 + 3c_2 &= 1 \\ c_2 &= 1 \end{aligned}$$

$$\begin{aligned} \text{for } c_1 \\ c_1 &= -2c_2 \\ c_1 &= -2 \end{aligned}$$

Thus,

$$e_1 \Rightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad ; \quad e_2 \Rightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix} = -2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Now by linearity of T :

$$\cancel{A = [T(e_1), T(e_2)]} \quad A = [T(e_1), T(e_2)]$$

$$\cancel{A = T\left(3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \end{bmatrix}\right)} \quad A = [T(3v_1 - v_2), T(-2v_1 + v_2)]$$

$$A = [3T(v_1) - T(v_2), -2T(v_1) + T(v_2)]$$

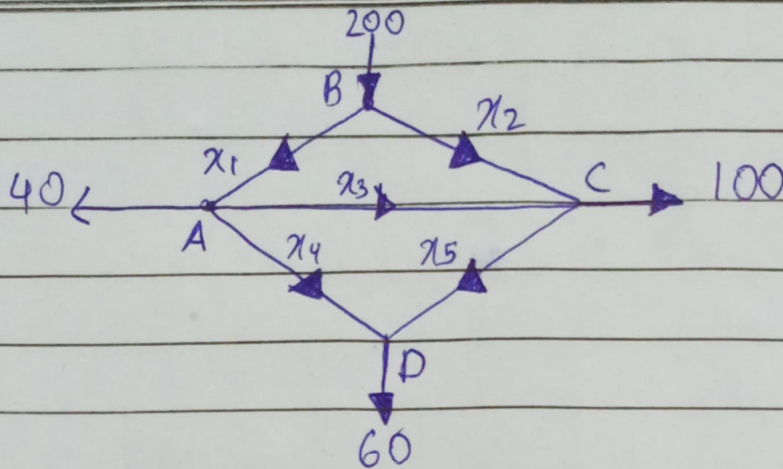
$$A = \left[3 \begin{bmatrix} 1 \\ -2 \end{bmatrix} - \begin{bmatrix} -2 \\ 5 \end{bmatrix}, -2 \begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} -2 \\ 5 \end{bmatrix} \right]$$

$$A = \left[\begin{bmatrix} 3+2 \\ -6-5 \end{bmatrix}, \begin{bmatrix} -2-2 \\ 4+5 \end{bmatrix} \right]$$

$$A = \left[\begin{bmatrix} 5 \\ -11 \end{bmatrix}, \begin{bmatrix} -4 \\ 9 \end{bmatrix} \right]$$

$$\therefore A = \begin{bmatrix} 5 & -4 \\ -11 & 9 \end{bmatrix} \quad \text{Answer}$$

Q13)



a)	Intersection	Flow in	Flow out
	A	x_1	$= x_3 + x_4 + 40$
	B	200	$= x_1 + x_2$
	C	$x_2 + x_3$	$= 100 + x_5$
	D	$x_4 + x_5$	$= 60$

Total Flow	200	=	200
------------	-----	---	-----

writing down in linear system format

$$x_1 - x_3 - x_4 = 40$$

$$x_1 + x_2 = 200$$

$$x_2 + x_3 - x_5 = 100$$

$$x_4 + x_5 = 60$$

Augmented Matrix

$$\left[\begin{array}{ccccc|c} 1 & 0 & -1 & -1 & 0 & 40 \\ 1 & 1 & 0 & 0 & 0 & 200 \\ 0 & 1 & 1 & 0 & -1 & 100 \\ 0 & 0 & 0 & 1 & 1 & 60 \end{array} \right]$$

Date: _____

$$\left[\begin{array}{ccccc|c} 1 & 0 & -1 & -1 & 0 & 40 \\ 0 & 1 & 1 & 1 & 0 & 160 \\ 0 & 1 & 1 & 0 & -1 & 100 \\ 0 & 0 & 0 & 1 & 1 & 60 \end{array} \right] \quad \begin{array}{l} \text{replacement} \\ R_2 - R_1 \end{array}$$

$$\left[\begin{array}{ccccc|c} 1 & 0 & -1 & -1 & 0 & 40 \\ 0 & 1 & 1 & 1 & 0 & 160 \\ 0 & 0 & 0 & -1 & -1 & -60 \\ 0 & 0 & 0 & 1 & 1 & 60 \end{array} \right] \quad \begin{array}{l} \text{replacement} \\ R_3 - R_2 \end{array}$$

$$\left[\begin{array}{ccccc|c} 1 & 0 & -1 & -1 & 0 & 40 \\ 0 & 1 & 1 & 1 & 0 & 160 \\ 0 & 0 & 0 & 1 & 1 & 60 \\ 0 & 0 & 0 & 1 & 1 & 60 \end{array} \right] \quad \begin{array}{l} \text{Scaling} \\ -R_3 \end{array}$$

$$\left[\begin{array}{ccccc|c} 1 & 0 & -1 & -1 & 0 & 40 \\ 0 & 1 & 1 & 1 & 0 & 160 \\ 0 & 0 & 0 & 1 & 1 & 60 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \quad \begin{array}{l} \text{replacement} \\ R_4 - R_3 \end{array}$$

$$\left[\begin{array}{ccccc|c} 1 & 0 & -1 & 0 & 1 & 100 \\ 0 & 1 & 1 & 0 & -1 & 100 \\ 0 & 0 & 0 & 1 & 1 & 60 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \quad \begin{array}{l} R_1 + R_3 \\ R_2 - R_3 \\ \text{replacement} \end{array}$$

So

$$x_1 - x_3 + x_5 = 100$$

$$x_3 = t \quad \text{where } t \in \mathbb{R}$$

$$x_2 + x_3 - x_5 = 100$$

$$x_5 = u \quad \text{where } u \in \mathbb{R}$$

$$x_4 + x_5 = 60$$

$$\therefore x_1 = 100 + t - u$$

$$x_3 = t \quad (\text{free})$$

$$x_2 = 100 - t + u$$

$$x_5 = u \quad (\text{free})$$

$$x_4 = 60 - u$$

b) $x_4 = 0$ as road is closed no cars can pass

so $x_4 = 60 - u$

$$0 = 60 - u$$

$$u = 60$$

hence $x_5 = 60$

as $x_5 = u$

$x_1 = 100 + t - u$	$x_2 = 100 - t + u$	$x_4 = 60 - u$	$x_3 = t$
$x_1 = 100 + t - 60$	$x_2 = 100 - t + 60$	$u = 60$	
$x_1 = 40 + t$	$x_2 = 160 - t$		

$\therefore$ when x_4 is closed

$$x_1 = 40 + t$$

$$x_2 = 160 - t$$

$$x_3 = t$$

$$x_4 = 0$$

$$x_5 = 60$$

so ~~$x_1 \geq 0$~~ ~~$x_2 \geq 0$~~

$$x_3 = t \geq 0$$

$$x_2 = 160 - t \geq 0$$

$$t \leq 160 \rightarrow \text{where } 0 \leq t \leq 160$$

~~$$0 \leq x_1 \leq 200$$~~

~~$$0 \leq x_2 \leq 160$$~~

c) when $x_4 = 0$ what is minimum x_1

from part b we get this

$$x_4 = 0 ; x_5 = 60 ; x_3 = t$$

$$x_2 = 160 - t ; x_1 = 40 + t$$

where $0 \leq t \leq 160$

put in eqⁿ

$$x_1 = 40 + 0$$

$$x_1 = 40$$

$$x_1 = 40 + 160$$

$$x_1 = 200$$

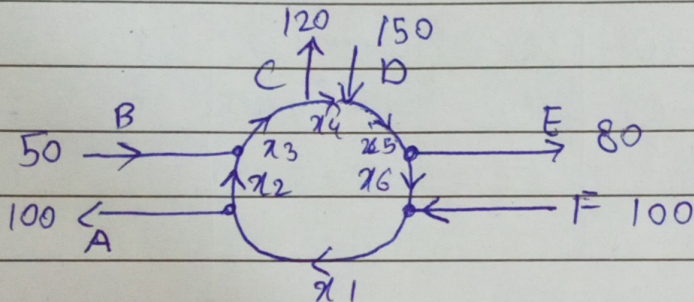
∴ minimum $x = 40$ dms

So for absolute ranges

$$x_1 \in [40, 200] \quad x_2 \in [0, 160] \quad x_3 \in [0, 160]$$

$$x_4 \in [0, 60] \quad x_5 \in [0, 60]$$

Q14)



Intersection	Flow in	=	Flow out
A	x_1	=	$100 + x_2$
B	$x_2 + 50$	=	x_3
C	x_3	=	$120 + x_4$
D	$x_4 + 150$	=	x_5
E	x_5	=	$80 + x_6$
F	$x_6 + 100$	=	x_1

Total Flow 300 = 300 consistent

Linear system

$$\begin{array}{rcl}
 \cancel{x_1} - 100 & x_1 - x_2 = 100 & A \\
 & x_2 - x_3 = -50 & B \\
 & x_4 - x_5 = -150 & D \\
 & x_3 - x_4 = 120 & C \\
 & x_5 - x_6 = 80 & E \\
 & x_6 - x_1 = -100 & F
 \end{array}$$

Augmented Matrix

$$\left[\begin{array}{cccccc|c} 1 & -1 & 0 & 0 & 0 & 0 & 100 \\ 0 & 1 & -1 & 0 & 0 & 0 & -50 \\ 0 & 0 & 1 & -1 & 0 & 0 & 120 \\ 0 & 0 & 0 & 1 & -1 & 0 & -150 \\ 0 & 0 & 0 & 0 & 1 & -1 & 80 \\ -1 & 0 & 0 & 0 & 0 & 1 & -100 \end{array} \right]$$

$$\left[\begin{array}{cccccc|c} \underline{1} & -1 & 0 & 0 & 0 & 0 & 100 \\ 0 & 1 & -1 & 0 & 0 & 0 & -50 \\ 0 & 0 & 1 & -1 & 0 & 0 & 120 \\ 0 & 0 & 0 & 1 & -1 & 0 & -150 \\ 0 & 0 & 0 & 0 & 1 & -1 & 80 \\ \cancel{0} & -1 & 0 & 0 & 0 & 1 & 0 \end{array} \right] \begin{array}{l} \text{replacement} \\ \\ \\ \\ \\ R_6 + R_1 \end{array}$$

$$\left[\begin{array}{cccccc|c} 1 & 0 & -1 & 0 & 0 & 0 & \cancel{100} 50 \\ 0 & \underline{1} & -1 & 0 & 0 & 0 & -50 \\ 0 & 0 & 1 & -1 & 0 & 0 & 120 \\ 0 & 0 & 0 & 1 & -1 & 0 & -150 \\ 0 & 0 & 0 & 0 & 1 & -1 & 80 \\ 0 & 0 & \underline{-1} & 0 & 0 & 1 & -50 \end{array} \right] \begin{array}{l} \text{replacement} \\ R_1 + R_2 \\ \\ \\ R_2 + R_6 \end{array}$$

$$\left[\begin{array}{cccccc|c} 1 & 0 & 0 & -1 & 0 & 0 & 170 \\ 0 & 1 & 0 & -1 & 0 & 0 & 70 \\ 0 & 0 & \underline{1} & -1 & 0 & 0 & 120 \\ 0 & 0 & 0 & 1 & -1 & 0 & -150 \\ 0 & 0 & 0 & 0 & 1 & -1 & 80 \\ 0 & 0 & 0 & -1 & 0 & 1 & 70 \end{array} \right] \begin{array}{l} \text{replacement} \\ R_1 + R_3 \\ R_2 + R_3 \\ \\ \\ R_6 + R_3 \end{array}$$

$$\left[\begin{array}{cccccc|c} 1 & 0 & 0 & 0 & -1 & 0 & 20 \\ 0 & 1 & 0 & 0 & -1 & 0 & -80 \\ 0 & 0 & 1 & 0 & -1 & 0 & -30 \\ 0 & 0 & 0 & \underline{1} & -1 & 0 & -150 \\ 0 & 0 & 0 & 0 & 1 & -1 & 80 \\ 0 & 0 & 0 & 0 & -1 & 1 & -80 \end{array} \right] \begin{array}{l} R_1 + R_4 \\ R_2 + R_4 \\ R_3 + R_4 \\ \\ R_5 + R_4 \\ R_6 + R_4 \end{array}$$

$$\left[\begin{array}{cccccc|c} 1 & 0 & 0 & 0 & 0 & -1 & 100 \\ 0 & 1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 50 \\ 0 & 0 & 0 & 1 & 0 & -1 & -70 \\ 0 & 0 & 0 & 0 & \underline{1} & -1 & 80 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{l} R_1 + R_5 \\ R_2 + R_5 \\ R_3 + R_5 \\ R_4 + R_5 \\ \\ R_6 + R_5 \end{array}$$

General Solution

$$x_1 - x_6 = 100$$

$$x_2 - x_6 = 0$$

$$x_3 - x_6 = 50$$

$$x_4 - x_6 = -70$$

$$x_5 - x_6 = 80$$

Let $t = x_6$ free parameter
where $t \in \mathbb{R}$

so $x_1 = 100 + t$

$$x_2 = t$$

$$x_3 = 50 + t$$

$$x_4 = -70 + t$$

$$x_5 = 80 + t$$

To find the smallest x_6 value

since ~~its~~ ~~on~~ the flow must be nonnegative each $x_i \geq 0$.

The smallest constrain can be found from

$$x_4 = -70 + t \geq 0$$

$$t \geq 70$$

Therefore the smallest physically feasible value of x_6 is

$$x_6^{\min} = 70$$

and this can be proven ($t = 70$)

$$x_1 = 100 + (70) = 170$$

$$x_2 = 70$$

$$x_3 = 50 + 70 = 120$$

$$x_4 = -70 + 70 = 0$$

$$x_5 = 80 + 70 = 150$$

$$x_6 = 70$$

} This proves our assumption that CD segment x_4 becomes zero at external point

Q15)

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (b-a)(c-a)(c-b)$$

$$\begin{bmatrix} 1 & 1 & 1 \\ a & b & c \\ 0 & b(b-a) & c(c-a) \end{bmatrix}$$

$$R_3 - aR_2$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & b-a & c-a \\ 0 & b(b-a) & c(c-a) \end{bmatrix}$$

$$R_2 - aR_1$$

Doing column determinant $[1, 0, 0]^T$

$$\begin{aligned}
 |\text{Determinant}| &= 1 \det \begin{bmatrix} b-a & c-a \\ b(b-a) & c(c-a) \end{bmatrix} \\
 &= \det \begin{bmatrix} b-a & c-a \\ b(b-a) & c(c-a) \end{bmatrix} \\
 &= (b-a)(c-a) - (c-a)(b(b-a)) \\
 &= (b-a)(c-a)
 \end{aligned}$$

Doing column determinant $[1, 0, 0]^T$

$$D = 1 \cdot \det \begin{bmatrix} b-a & c-a \\ b(b-a) & c(c-a) \end{bmatrix}$$

$$D = (b-a)(c-a) \det \begin{bmatrix} 1 & 1 \\ b & c \end{bmatrix}$$

$$D = (b-a)(c-a)(c-b)$$

hence shown